

1a	point D (vertical speed first starts to decrease)
1bi	$\text{Deceleration} = \frac{(50 - 20)}{2}$ $= 15 \text{ m s}^{-2}$
1bii	$s = ut + \frac{1}{2} a t^2$ $= 50(2) + \frac{1}{2}(-15)(2)^2$ $= 70 \text{ m}$ Alternative method is calculate area under graph $= \frac{1}{2}(50+20)(2)$ $= 70 \text{ m}$
1biii	$F_{\text{net}} = ma$ $mg - F_R = ma$ $85 \times 9.81 - F_R = 85(-15)$ $F_R = 2100 \text{ N}$
1c	<u>Air resistance is larger than weight</u>, hence resultant force points upwards but velocity is downwards, hence parachutist slows down. <u>Air resistance decreases (as speed decreases)</u>, hence resultant force decreases to 0 After which, <u>air resistance equals to the weight</u> at point F and hence velocity remains constant.
1di	<u>The parachutist comes to a complete stop over a longer period (for a given change in momentum, with a longer period, this) reducing the force of impact.</u> <u>The force of impact on the parachutist is also distributed across various parts of the body (instead of a single part), reducing the risk of injury.</u>
1dii	Considering the parachutist only as a system, As there is net external force by Earth on parachutist, momentum is not conserved. Or Considering the parachutist and Earth as a system, As there is no external force acting on the system, momentum is conserved as there is a corresponding change in the momentum of the Earth.
2ai	Normal contact (reaction) force towards centre of circular motion